Report C1

Reasoning

I defined < and <= in for the Rectangle. Firstly, we need to compare the area of the Rectangle, but the compiler could not know its semantic meaning. Therefore, I defined < as an approach to let the compiler know what the actual comparison is. Meanwhile, I defined <= as well, since in the merge_sort, the circumstance when the left_half[left_half_index] == right_half[right_half_index] is included in the else.

```
while (left_half_index < left_have_size && right_half_index < right_have_size)
{
    if (left_half[left_half_index] < right_half[right_half_index])
    {
        arr[result_index] = left_half[left_half_index];
        left_half_index++;
    }else
    {
        arr[result_index] = right_half[right_half_index];
        right_half_index++;
    }
    result_index++;
}
```

That means the when the left one equals to the right one, we would choose the right one priorly. In this case, <= is used to ensure the right one is always switched into left.

Evidence of Testing

Test	Screen Shoot
generic_sort.cpp	/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/C1/generic_sort Everything seems fine :) Process finished with exit code 0
generic_linked_list.cpp int test	/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/C1/generic_linked_list Linked List: 10 20 30 Linked List after setting index 1 to 25: 10 25 30 Item at index 2: 30
generic_linked_list.cpp string test	Linked List: Apple Banana Cat Linked List after setting index 1 to Peach: Apple Peach Cat Item at index 2: Cat
generic_linked_list.cpp double test	Linked List: 10 11 12 Linked List after setting index 1 to 11.5: 10 11.5 1 Item at index 2: 12